

Massless Scalars in 2D:

(i) Have D-free scalar fields in 2D
 $x^\mu(\sigma^1, \sigma^2)$.

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma (\partial_1 x^\mu \partial_1 x_\mu + \partial_2 x^\mu \partial_2 x_\mu)$$

Polyakov in $\mathbb{E}^D$. Flat worldsheet $\delta_{\alpha\beta} : (+1)$

* Let us define Complex coordinates $\Sigma_g \simeq \mathbb{C}$

$$z = \sigma^1 + i\sigma^2, \quad \bar{z} = \sigma^1 - i\sigma^2$$

$$\partial_z = \frac{1}{2}(\partial_1 - i\partial_2), \quad \partial_{\bar{z}} = \frac{1}{2}(\partial_1 + i\partial_2)$$

$$\partial = \partial_z, \quad \bar{\partial} = \partial_{\bar{z}}$$

$$V^z = V^1 + iV^2, \quad V^{\bar{z}} = V^1 - iV^2$$

$$V_z = \frac{1}{2}(V^1 - iV^2), \quad V_{\bar{z}} = \frac{1}{2}(V^1 + iV^2)$$

$$[g_{\alpha\beta}]^c = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, \quad [g^{\alpha\beta}]^c = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$$

$$g_{zz} = g_{\bar{z}\bar{z}} = 0, \quad g_{z\bar{z}} = g_{\bar{z}z} = \frac{1}{2}$$

$$d\bar{z}^2 = 2 d\sigma^1 d\sigma^2$$

$$d\bar{z} = \left| \frac{\partial \bar{z}}{\partial \sigma} \right| d\sigma^1 d\sigma^2 \Rightarrow d^2 z |g|^{-1/2} = d\sigma^1 d\sigma^2$$

Delta δ^u : $\int d^2 z \delta^2(z, \bar{z}) = 1$

$$\int d^2 z \delta^2(z, \bar{z}) = \int d\sigma^1 d\sigma^2 2 \delta(z) \delta(\bar{z})$$

$$\delta(f(x)) = \delta(x) / \sum_{x_*} f'(x_*) =$$

$$\therefore \delta^2(z, \bar{z}) = \frac{1}{2} \delta(\sigma^1) \delta(\sigma^2)$$

Divergence thm: $\int_{\mathbb{R}} d^2 z (\partial_z V^z + \partial_{\bar{z}} V^{\bar{z}})$

$$= i \oint_{\partial \mathbb{R}} (V^z d\bar{z} - V^{\bar{z}} dz)$$

$\leftarrow \partial \mathbb{R}$

Proof:

$$\int_{\mathbb{R}} dz d\bar{z} \partial_z v^z + dz d\bar{z} \partial_{\bar{z}} v^{\bar{z}} \\ = \int_{\mathbb{R}} [d\bar{z} (\partial v^z) + dz \partial v^{\bar{z}}]$$

i and (-) come from going to $\sigma^1 \sigma^2$ and using Stokes.

* $S = \frac{1}{2\pi\alpha'} \int d^2z \partial X^\mu \bar{\partial} X_\mu$

E.O.M: $\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \partial_\mu X^\sigma} \right) - \frac{\partial \mathcal{L}}{\partial X^\sigma} = 0$

$$\partial \left(\frac{\partial \mathcal{L}}{\partial \partial X^\sigma} \right) + \bar{\partial} \frac{\partial \mathcal{L}}{\partial \bar{\partial} X^\sigma} = 0$$

$$\partial \bar{\partial} X^\sigma + \bar{\partial} \partial X^\sigma = 0 \Rightarrow \partial \bar{\partial} X^\mu(z, \bar{z}) = 0$$

$$\bar{\partial} \partial X^\mu(z, \bar{z}) = 0$$

Note: $\partial(\bar{\partial} X^\mu) = \bar{\partial}(\partial X^\mu) = 0$

∂X^μ : Holomorphic

$\bar{\partial} X^\mu$: Anti-holo

$$\Rightarrow \partial X^\mu = \partial X^\mu(z)$$

$$\bar{\partial} X^\mu = \bar{\partial} X^\mu(\bar{z})$$

Wick rotate: $\sigma^2 \rightarrow i\sigma^0$

$$f(z) = f(\sigma^1 - \sigma^2)$$

$$f(\bar{z}) = f(\sigma^1 + \sigma^2)$$

Holomorphic: Left-moving

Anti holo: Right-moving

* Expectation values:

$\mathcal{F}[X]$: is any functional of X

$$\langle \mathcal{F}[X] \rangle = \int [dX] \exp(-S) \mathcal{F}[X]$$

• Integral over X^0 has a wrong Gaussian sign: Understood via a Wick-rotation
 $X^0 \mapsto -iX^0$
 $\Rightarrow \langle \rangle$ is not normalized.

* Path int. of $\frac{\delta}{\delta X}$ total derivative is zero

$$0 = \int [DX] \frac{\delta}{\delta X_\mu(z, \bar{z})} e^{-S} = - \int DX e^{-S} \frac{\delta S}{\delta X_\mu(z, \bar{z})}$$

$$= - \left\langle \frac{\delta S}{\delta X_\mu(z, \bar{z})} \right\rangle = \frac{1}{4\pi\alpha'} \left\langle \partial \bar{\partial} X^\mu(z, \bar{z}) \right\rangle$$

Can add insertions '...' so long as they are not evaluated at z .

$$\left\langle \partial \bar{\partial} X^\mu(z, \bar{z}) \dots \right\rangle = 0 \quad (\text{operator Equations})$$

Insertions can be seen as $|in\rangle$ and $\langle out|$ states

• In the Hilbert space of states:

$$\partial \bar{\partial} \hat{X}^\mu(z, \bar{z}) = 0$$

* If we consider cases w/ ops inserted at z : then

$$0 = \int DX \frac{\delta}{\delta X_\mu(z, \bar{z})} \left[e^{-S} X^\nu(z', \bar{z}') \right]$$

$$= \int DX \left(- \frac{\delta S}{\delta X_\mu(z, \bar{z})} X^\nu + \eta^{\mu\nu} \delta(z - z') \delta(\bar{z} - \bar{z}') \right) e^{-S}$$

$$= \int DX \left(\frac{1}{4\pi\alpha'} (\partial_z \bar{\partial}_{\bar{z}} X^\mu(z, \bar{z})) X^\nu(z', \bar{z}') + \eta^{\mu\nu} \delta^2(z - z', \bar{z} - \bar{z}') \right) e^{-S}$$

$$= \eta^{\mu\nu} \langle \delta^2(z - z', \bar{z} - \bar{z}') \rangle + \frac{1}{4\pi\alpha'} \partial_z \bar{\partial}_{\bar{z}} \langle X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') \rangle$$

• F.O.M holds except for at coincident pts.

$\Rightarrow$ Additional insertions must not be at

$$\frac{1}{\pi\alpha'} \partial_z \partial_{\bar{z}} \langle X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') \dots \rangle = \eta^{\mu\nu} \langle \delta^2(z-z', \bar{z}-\bar{z}') \rangle$$

which, at the level of operators:

$$\frac{1}{\pi\alpha'} \partial_z \partial_{\bar{z}} X^\mu(z, \bar{z}) X^\nu(z', \bar{z}') = -\eta^{\mu\nu} \delta^2(z-z', \bar{z}-\bar{z}')$$

Normal ordering: "move annihilation to Right"

$$:a^\dagger a a^\dagger: = a^\dagger a^\dagger a$$

$$T(\phi_1 \dots \phi_n) = : \phi_1 \phi_2 : + \dots \text{ normal ordering}$$

$$T(\phi_1 \phi_2) = : \phi_1 \phi_2 : + \overbrace{\phi_1 \phi_2}$$

$$T(\phi_1 \phi_2) - \phi_1 \phi_2 = : \phi_1 \phi_2 :$$

$$\int \frac{d^4 p e^{i p \cdot (p_1 - p_2)}}{(p_1 - p_2)^2 + m^2 + i\epsilon}$$

$$:X^\mu(z_1, \bar{z}_1) X^\nu(z_2, \bar{z}_2): = X^\mu X^\nu + \frac{\alpha'}{2} \eta^{\mu\nu} \ln |z_{12}|^2$$

$$z_{ij} \equiv z_i - z_j$$

$$\partial_z \partial_{\bar{z}} :X^\mu(z_1, \bar{z}_1) X^\nu(z_2, \bar{z}_2): = 0$$

$$\partial_z \partial_{\bar{z}} X^\mu X^\nu + \frac{\alpha'}{2} \partial_z \partial_{\bar{z}} \ln |z_{12}|^2$$

$$|z_{12}|^2 = z_{12} \bar{z}_{12} = (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$

$$\therefore \boxed{\partial_z \partial_{\bar{z}} :X^\mu_{11} X^\nu_{22}: = 0}$$

$$\partial \bar{\partial} \ln |z|^2 = 2\pi \delta^2(z, \bar{z})$$

$$\ln |z|^2 = \ln(z) + \ln(\bar{z})$$



The operator product expansion: OPE

want to understand

$$\langle \mathcal{O}_{i_1}(z_1, \bar{z}_1) \mathcal{O}_{i_2}(z_2, \bar{z}_2) \dots \mathcal{O}_{i_n}(z_n, \bar{z}_n) \rangle$$

where $(z_i, \bar{z}_i) \rightarrow (z_j, \bar{z}_j)$ coincide.

* Here we will take $z_1 \rightarrow z_2$

with the radius of convergence being the distance

pt of the nearest operator



$$\phi_i(\sigma_1) \phi_j(\sigma_2) = \sum_k C_{ij}^k(\sigma_1 - \sigma_2) \phi_k(\sigma_2)$$

$$\Leftrightarrow \langle \phi_i(\sigma_1) \phi_j(\sigma_2) \dots \rangle = \sum_k C_{ij}^k(\sigma_1 - \sigma_2) \langle \phi_k(\sigma_2) \dots \rangle$$

coeff. $f_{ij}^k: C_{ij}^k$.

$\rightarrow$ Dependence of other ops comes from exp. values on the R.H.S.

$\partial_{\bar{z}} : X^\mu X^\nu : = 0 \Rightarrow : X^\mu X^\nu :$ is Harmonic
so it is locally a sum of a holo and anti holo f^a :

$$X^\mu(z_1, \bar{z}_1) X^\nu(z_2, \bar{z}_2) = -\frac{\alpha'}{2} \eta^{\mu\nu} \ln |z_{12}|^2 + :X^\mu X^\nu(z_2, \bar{z}_2):$$

$$+ \sum_{k=1}^{\infty} \frac{1}{k!} \left[(z_{12})^\mu : X^\nu \partial^\mu X^\mu(z_2, \bar{z}_2) : + (\bar{z}_{12})^\nu : \bar{X} \bar{\partial}^\nu X^\mu : \right]$$

Terms w/ mixed $\partial \bar{\partial}$ vanish by $e\Phi M$

• By letting $a' = z$

$$= -\eta^{\mu\nu} \ln |z_{12}|^2 + \dots$$

$\rightarrow$ Common in the literature.

* Gives the behavior as $z_1 \rightarrow z_2$ as an ∞ series

Normal ordering:

$$: X^{\mu_1}(z_1, \bar{z}_1) \dots X^{\mu_n}(z_n, \bar{z}_n) : = X^{\mu_1}(z_1, \bar{z}_1) X^{\mu_n}(z_n, \bar{z}_n)$$

+ $\sum$ Subtractions

$\rightarrow$ all ways of choosing one, two, or more

replacing the pair w/ $\frac{1}{2} \alpha' \eta^{\mu\nu} k_\mu k_\nu |z_{ij}|^2$

Aside: Scalar field:

$$S = \frac{1}{2} \int d^2x \sqrt{-g} \eta^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi$$

$$= \frac{1}{2} \int d^2x (\partial_1 \phi)^2 + (\partial_2 \phi)^2$$

$$= \frac{1}{2} \int d^2z d^2\bar{z} \partial \phi \cdot \bar{\partial} \phi$$

E.O.M $\partial \bar{\partial} \phi(z, \bar{z}) = 0 \Rightarrow \phi(z, \bar{z}) = \phi(z) + \phi(\bar{z})$

$\Rightarrow G(z_1, \bar{z}_1, z_2, \bar{z}_2) = \langle \phi(z_1, \bar{z}_1) \phi(z_2, \bar{z}_2) \rangle$

$\therefore \phi(z_1, \bar{z}_1) = \int d^2z_2 d^2\bar{z}_2 G_{1,22} \phi_{22}$

$\partial \bar{\partial} G_{1,22} = -\frac{1}{g} \delta^{(2)}(z_1 - z_2)$

where $\partial \bar{\partial} \ln|z_1 - z_2|^2 = \partial \left(\frac{1}{z_1 - z_2} \right) = 2\pi \delta^{(2)}(z_1 - z_2)$

$:X^{\mu_1} X^{\mu_2} X^{\mu_3}: = X^{\mu_1} X^{\mu_2} X^{\mu_3} + \overbrace{X^{\mu_1} X^{\mu_2} X^{\mu_3}} + \overbrace{X^{\mu_1} X^{\mu_2} X^{\mu_3}} + \overbrace{X^{\mu_1} X^{\mu_2} X^{\mu_3}}$

The 'wick' identity can be summarized as follows:

$:K: = \exp \left(\frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \frac{\delta}{\delta X^{\mu_1}} \frac{\delta}{\delta X^{\mu_2}} \right) P$

where $P = P[X]$ is a functional of X^μ

Proof:

$:X^{\mu_1} X^{\mu_2} X^{\mu_3}: = \exp \left(\frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \frac{\delta}{\delta X^{\mu_1}} \frac{\delta}{\delta X^{\mu_2}} \right) (X^{\mu_1} X^{\mu_2} X^{\mu_3})$

$$= X^{\mu_1} X^{\mu_2} X^{\mu_3} + \frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \left(\frac{\delta}{\delta X^{\mu_1}} \frac{\delta}{\delta X^{\mu_2}} \right) (X^{\mu_1} X^{\mu_2} X^{\mu_3})$$

$$= X^{\mu_1} X^{\mu_2} X^{\mu_3} + \frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \delta(z_2 - \omega_1) \delta^{\mu_1} \frac{\delta}{\delta X^{\mu_1}} (X^{\mu_1} X^{\mu_2} X^{\mu_3})$$

$$= X^{\mu_1} X^{\mu_2} X^{\mu_3} + \frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \delta(z_2 - \omega_1) \delta^{\mu_1} X^{\mu_2} X^{\mu_3}$$

$$+ \dots$$

$$= X^{\mu_1} X^{\mu_2} X^{\mu_3} + \frac{\alpha'}{4} \int d^2z_1 d^2z_2 \ln|z_{12}|^2 \delta(z_2 - \omega_1) \delta^{\mu_1} X^{\mu_2} X^{\mu_3} + \dots$$

$$= X^{\mu_1} X^{\mu_2} X^{\mu_3} + \frac{\alpha'}{2} \ln |z_{12}|^2 M^{\mu_1 \mu_2} X^{\mu_3} + \text{perm} \quad \square$$

Applying the inverse:

$$F = \exp\left(-\frac{\alpha'}{4} \int \frac{\partial^2}{\partial X^{\mu} \partial X^{\nu}}\right) :F:$$

$$= :F: + \text{contractions}$$

→ Contractions, opposite of subtraction

↳ Def: Sum over all ways of choosing one, two, or more pairs of fields from $:F:$ and replacing the pair w/

$$-\frac{\alpha'}{2} M^{\mu\nu} \ln |z_{ij}|^2$$

OPE:

$$:F:(z) :G:(z') = e^{\left(\frac{\alpha'}{4} \int d^2 z_1 d^2 z_2 \ln |z_{12}|^2 \frac{\partial}{\partial X_1^{\mu}} \frac{\partial}{\partial X_{\mu 2}}\right)} \\ \times e^{\left(\frac{\alpha'}{4} \int d^2 z_3 d^2 z_4 \ln |z_{34}|^2 \frac{\partial}{\partial X_3^{\nu}} \frac{\partial}{\partial X_{\nu 4}}\right)} F G \\ = \exp\left(z_1 + z_2 + \left(\frac{\alpha'}{4}\right)^2 \left(\prod_{i=1}^4 d^2 z_i\right) \left[\ln |z_{12}|^2 \frac{\partial}{\partial X_1^{\mu}} \frac{\partial}{\partial X_{\mu 2}} \frac{\partial}{\partial X_3^{\nu}} \frac{\partial}{\partial X_{\nu 4}}\right]\right) \\ = \exp(\Sigma)$$

Not sure how we obtain $:FG:$ since we have a sum over two different z 's in $\exp(\int d^2 z_1 d^2 z_2 \dots)$.

$$:F::G: = :FG: + \Sigma \text{ cross-contractions}$$

$$F = F[X], \quad G = G[X]$$

or

$$:F::G: = \exp\left(-\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 \ln |z_{12}|^2 \frac{\partial}{\partial X_F^{\mu}} \frac{\partial}{\partial X_{\mu G}}\right) \\ \times :FG:$$

$$\text{ex: } :\partial X^{\mu}(z) \partial X_{\mu}(z)::\partial' X^{\nu}(z') \partial' X_{\nu}(z'):$$

$$= :\partial X^{\mu} \partial X_{\mu} \partial' X^{\nu} \partial' X_{\nu}: + \text{cross contractions}$$

$$\begin{aligned}
& = \underbrace{\partial X^m \partial X_n \partial^2 X^m \partial^2 X^m}_{\text{Double contractions}} + \underbrace{\partial X^m \partial^2 X^m \partial^2 X^m}_{\text{Double contractions}} \\
& = 4 \partial X^m \partial^2 X^m \partial X_n \partial^2 X_n + \dots \\
& = 4 \left(-\frac{\alpha'}{2} \partial \partial^2 \eta^{mn} \ln |z - z'|^2 \right) \partial X_n \partial^2 X^m + \dots \\
& = -\frac{4\alpha'}{2} \partial \partial^2 \ln |z - z'|^2 \partial X_n \partial^2 X^m + \dots \\
& = \underline{\underline{(\dots)}} + \underline{\underline{2 \eta^{mn} \left(-\frac{\alpha'}{2} \partial \partial^2 \ln |z - z'|^2 \right)^2}}
\end{aligned}$$

Recall that Taylor expanding we have

$$\begin{aligned}
X^m_1, X^m_{22} &= -\frac{\alpha'}{2} \eta^{mn} \ln |z_1, z_2|^2 + :X^m X^m(z_2, \bar{z}_2): \\
&+ \sum_{k=1}^{\infty} \frac{1}{k!} (z_1)^k :X^m \partial^k X^m_{22}: + (\bar{z}_1)^k :X^m \bar{\partial}^k X^m_{22}:
\end{aligned}$$

and

$$\begin{aligned}
\partial \ln |z - z'|^2 &= \partial \left(\frac{1}{|z - z'|^2} |z - z'| \right) \\
&= \partial \left(\frac{1}{z - z'} \right) = \frac{1}{(z - z')^2}
\end{aligned}$$

* Note this is $(z - z')^2$ NOT $|z - z'|^2$
 where $|z - z'|^2 = (z - z')(\bar{z} - \bar{z}')$

$$\therefore \eta^{mn} = \mathbb{D}$$

$$\rightarrow \frac{\partial \alpha'}{(2(z - z'))^4}$$

$$\rightarrow -\frac{2\alpha'}{(z - z')^2} : \partial X_m(z) \partial^2 X^m(z') :$$

Not sure about $z \rightarrow z'$

$$\begin{aligned}
: \partial X_m(z) \partial^2 X^m(z') : &= : -\frac{\alpha'}{2} \eta^{mn} : + : z_2 : \\
&+ z_2 : \partial^2 X_m \partial^2 X^m : + \bar{z}_2
\end{aligned}$$

(We are assuming X^m is holo).

$$\text{ex: } \mathcal{L} = i k_1 \cdot \dot{X}(z, \bar{z})$$

$$\mathcal{L} = i k_2 \cdot \dot{X}(0, 0)$$

$$\Rightarrow \frac{\partial}{\partial x_F^\mu} F = i k_1^\mu F, \quad \frac{\partial}{\partial x_g^\mu} G = i k_2^\mu G$$

$$\begin{aligned} \therefore K : : \phi : &= \exp \left(-\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 h(z_1, z_2) \frac{\delta}{\delta x_F^\mu} \frac{\delta}{\delta x_{g\mu}} \right) : F G : \\ &= \sum_{n=0}^{\infty} \left(-\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 h(z_1, z_2) \frac{\delta}{\delta x_F^\mu} \frac{\delta}{\delta x_{g\mu}} \right)^n : F G : \\ &= \sum_{n=0}^{\infty} \left(-\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 (i k_{1\mu}) (i k_2^\mu) \right)^n : e^{i k_1 x} e^{i k_2 x_{0,0}} : \\ &= \exp \left(\frac{\alpha'}{2} \int d^2 z_1 d^2 z_2 \delta^2(z_1 - z) \delta^2(z_2 - 0) (k_1, k_2^\mu) \right) \\ &= \exp \left(\frac{\alpha'}{2} h(z_1, z_2) k_{1\mu} k_2^\mu \right) : e^{i k_1 \cdot X(z, \bar{z})} e^{i k_2 \cdot X(0,0)} : \\ &= |z|^{\alpha'(k_1, k_2)} : e^{i k_1 \cdot X(z, \bar{z})} e^{i k_2 \cdot X(0,0)} : \end{aligned}$$

Taylor expanding

$$e^{i k_1 \cdot X(z, \bar{z})} = e^{i k_1 \cdot [X(0,0) + z \partial X + \dots]}$$

so, via Baker:

$$|z|^{\alpha'(k_1, k_2)} : e^{(i k_1 + i k_2) \cdot X} + \dots$$

$$\therefore K : : \phi : = |z|^{\alpha'(k_1, k_2)} : e^{i(k_1 + k_2) \cdot X(0,0)} (1 + \mathcal{O}(z, \bar{z})) :$$

Ward Identities and Noether's theorem:

* Consider a general field theory with action $S[\phi]$ in d -dim.

$\phi_\alpha(\sigma)$: General field of spin s .

$$\Rightarrow \phi'_\alpha(\sigma) = \phi_\alpha(\sigma) + \delta \phi_\alpha(\sigma)$$

$$\delta \phi_\alpha(\sigma) := \epsilon \Delta \phi_\alpha$$

In the path integral:

$$D\phi' e^{-S[\phi']} \text{ is invariant}$$

Proof:

$$\begin{aligned} D\phi' &= \lim_{N \rightarrow \infty} \left(\prod_{\alpha=1}^N \int d\phi'_\alpha \right) \\ &= \lim_{N \rightarrow \infty} \left(\prod_{\alpha=1}^N \int d\phi_\alpha \det \left(\frac{\partial \phi'_\alpha}{\partial \phi_\alpha} \right) \right) \\ S[\phi'] &= S[\phi] + \delta \phi S'[\phi] + \dots \end{aligned}$$

$$\prod_{i=1}^n \exp(S[\phi_i]) \prod_{i=1}^n \exp(S[\phi_i])$$

* A continuous symm $\Rightarrow$ A conserved Noether current as well as Ward Identities

$\rightarrow$ Let us now consider the transformation:

$$\phi'_\alpha(\sigma) = \phi_\alpha(\sigma) + \rho(\sigma) \delta \phi_\alpha(\sigma)$$

$$\phi'_\alpha = \mathcal{U} \phi_\alpha \Rightarrow D\phi' = \det(\mathcal{U}) D\phi$$

$$\det(e^{\rho(\sigma) \delta \phi(\sigma)}) = \exp(\text{Tr}(\log(\rho(\sigma) \delta \phi)))$$

$$= \exp(\text{Tr}(\rho(\sigma) \delta \phi(\sigma)))$$

$$= \exp(\text{Tr}(\rho \delta \phi)) e^{-S[\phi]} \prod_{n=1}^{\infty} e^{(\rho \delta \phi)^n S^{(n)}[\phi]}$$

$$= \mathcal{D}\phi e^{-S[\phi]} \exp(\text{Tr}(\rho(\sigma) \delta \phi(\sigma))) \prod_{n=1}^{\infty} e^{(\rho(\sigma) \delta \phi(\sigma))^n S^{(n)}[\phi]}$$

$$\rightarrow D\phi' e^{-S[\phi']} = D\phi e^{-S[\phi]} \left(1 + \frac{i\epsilon}{2\pi} \int d^d \sigma g^{1/2} \nabla_a j^a \rho(\sigma) + \mathcal{O}(\epsilon^2) \right)$$

Take $\rho(\sigma)$ to be non-zero in a ^{small} region $\mathcal{U}$. With insertions $\mathcal{O}_i$ outside of $\mathcal{U}$



Insertions: ...

$$\therefore \mathcal{O}_i(\tilde{\sigma}) = \mathcal{O}_i(\sigma)$$

$$\text{since } \rho(\tilde{\sigma}) = 0 \quad \tilde{\sigma} \in \mathcal{U}$$

* Invariance of the path integral under ϕ' :

$$\begin{aligned} \Rightarrow 0 &= \int D\phi' e^{-S[\phi']} - \int D\phi e^{-S[\phi]} \\ &= \int D\phi e^{-S} \left\{ \frac{\epsilon}{2\pi} \int d^d \sigma g^{1/2} \nabla_a j^a \rho(\sigma) \right\} \dots \\ &= \frac{\epsilon}{2\pi} \int d^d \sigma g^{1/2} \rho(\sigma) \langle \nabla_a j^a(\sigma) \dots \rangle \end{aligned}$$

$$11 \dots 17:8$$

expression.

For the Ward Identity: Let $p(\sigma)$ be a test f^u in some region

$$R \quad p(\sigma) = \begin{cases} 1, & \sigma \in R \subset \mathcal{U} \\ 0, & \sigma \notin R \end{cases}$$

Let $\sigma_0 \in R$. And insert $\mathcal{A}(\sigma_0)$

Then $S[\phi]$

$$\begin{aligned} & \int D\phi' e^{-S[\phi']} \mathcal{A}(\sigma_0) \dots \quad \leftarrow p(\sigma_0) = 1 \\ &= \int D\phi e^{-S[\phi]} (\mathcal{A}(\sigma_0) + \delta \mathcal{A}(\sigma_0)) (1 + \epsilon) \\ &= \int D\phi e^{-S[\phi]} (\mathcal{A}(\sigma_0) \dots + \delta \mathcal{A}) \\ &\quad + \int D\phi e^{-S[\phi]} (\mathcal{A} + \delta \mathcal{A}) \epsilon \\ &= \int D\phi e^{-S} \mathcal{A}(\sigma_0) \dots + \int D\phi e^{-S} \delta \mathcal{A}(\sigma_0) \dots \\ &\quad + \int D\phi e^{-S} (\mathcal{A} + \delta \mathcal{A})(\sigma_0) \left(\frac{\epsilon}{2\pi i} \int d^2\sigma \nabla_a j^a p(\sigma) g^{1/2} \right) \\ &\stackrel{!}{=} \int D\phi e^{-S} \mathcal{A}(\sigma_0) \dots \\ &\quad \langle \delta \mathcal{A} \dots \rangle + \langle (\mathcal{A} + \delta \mathcal{A}) \frac{\epsilon}{2\pi i} \int_R d^2\sigma \nabla_a j^a \end{aligned}$$

Note $\delta \mathcal{A} = \delta \phi_a \frac{\partial \mathcal{A}}{\partial \phi_a} |_{\phi' = \phi}$

$$\therefore \delta \mathcal{A} \left(\frac{\epsilon}{2\pi i} \right) \rightarrow 0$$

$$\langle \delta \mathcal{A} \dots \rangle + \left\langle \left(\frac{\epsilon}{2\pi i} \int d^2\sigma g^{1/2} \nabla_a j^a(\sigma) \right) \mathcal{A}(\sigma_0) \right\rangle = 0$$

So, classically:

$$\delta \mathcal{A}(\sigma_0) + \frac{\epsilon}{2\pi i} \int d^2\sigma g^{1/2} \nabla_a j^a(\sigma) \mathcal{A}(\sigma_0) = 0$$

Note: $\delta \mathcal{A}(\sigma_0) = \int d^2\sigma \delta^d(\sigma - \sigma_0) \delta \mathcal{A}(\sigma)$

$$\Rightarrow \int d\sigma \left[\partial_- (\sigma - \sigma_0) \delta A(\sigma) + \frac{2\pi i g}{\epsilon} V_{aj}(\sigma) A(\sigma_0) \right] = 0$$

$$\underbrace{\qquad\qquad\qquad}_{\text{divergence term}} + \frac{2\pi i g}{\epsilon} V_{aj}(\sigma) A(\sigma_0) + \frac{d}{d\sigma}(\dots) = 0$$

Via the divergence theorem:

$$\int_R d\sigma \partial_{aj}(\sigma) A(\sigma_0) = \int_{\partial R} dA n_{\mu} j^{\mu}(\sigma) A(\sigma_0)$$

$$\underbrace{\int_{\partial R} dA n_{\mu} j^{\mu}(\sigma) A(\sigma_0)}_{\text{divergence term}} = \frac{2\pi i g}{\epsilon} \delta A(\sigma_0)$$

Let $d=2$ and $R \subset \mathbb{C}$

$$\therefore dA = dz d\bar{z}$$

$$\sigma_0 = (z_0, \bar{z}_0)$$

$$\text{Recall } dA = i(\cdot) d\bar{z} - (\cdot) dz$$

$$i \oint_R (i dz + \tilde{j} d\bar{z}) = \frac{2\pi i g}{\epsilon} \delta A(z_0, \bar{z}_0)$$

$$\oint_{\partial R} (i dz - \tilde{j} d\bar{z}) = \frac{2\pi i g}{\epsilon} \delta A(z_0, \bar{z}_0)$$

$$j \equiv j_z, \quad \tilde{j} \equiv j_{\bar{z}}$$

j_z : holo | $\tilde{j}_{\bar{z}}$: Anti-holo.

and are separately conserved:

$$\boxed{\begin{aligned} & \text{Res}_{z \rightarrow z_0} (j(z) A(z_0, \bar{z}_0)) + \overline{\text{Res}}_{\bar{z} \rightarrow \bar{z}_0} [\tilde{j}(\bar{z}) A(z_0, \bar{z}_0)] \\ &= \frac{i g}{\epsilon} \delta A(z_0, \bar{z}_0) \end{aligned}}$$

ex. Massless scalar:

$$\text{Spacetime translations } \delta X^{\mu} = \epsilon a^{\mu} \rho(\sigma)$$

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \eta^{\alpha\beta} \partial_{\alpha} X^{\mu} \partial_{\beta} X_{\mu}$$

$$\delta S = \frac{1}{2\pi\alpha'} \int d^2\sigma \partial_{\alpha} X^{\mu} \partial_{\alpha} (\epsilon a_{\mu} \rho(\sigma))$$

Recall that:

$$\frac{1}{2\pi i} \oint d\sigma g^{1/2} p(\sigma) \langle \nabla_a j^a(\sigma) \dots \rangle = 0$$

$\therefore$

$Q_{\mu} j^{\mu}_a = \text{conserved current}$

$$j^{\mu}_a = \frac{i}{\alpha'} \partial_a x^{\mu}$$

$$j^{\mu}(z) : e^{ikX(0,0)} : \sim \frac{K^{\mu}}{2z} : e^{ikX(0,0)} :$$

$$\begin{aligned} \text{proof: } & : \frac{i}{\alpha'} \partial_a x^{\mu} : : e^{ikX(0,0)} : \\ &= \frac{i}{\alpha'} : \partial_a x^{\mu} e^{ikX(0,0)} : + : \partial_a x^{\mu} \sum_{n=0}^{\infty} \frac{1}{n!} (ik \cdot X_0)^n : \\ &= \frac{i}{\alpha'} : \partial_a x^{\mu} e^{ikX(0,0)} : + \partial_z x^{\mu} ik^{\mu} X(0,0) + \dots \\ &= \frac{i}{\alpha'} : \partial_a x^{\mu} e^{ikX(0,0)} : + \partial_z K_{\mu} \eta^{\mu\nu} \ln|z|^2 \left(-\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} & \frac{i}{\alpha'} : \partial_a x^{\mu} e^{ikX(0,0)} : \partial_z \ln|z|^2 K^{\mu} \left(\pi \frac{1}{2}\right) \alpha' + \dots \\ &= \frac{i}{2\alpha'} : \partial_a x^{\mu} e^{ikX(0,0)} : + \left(\frac{1}{2}\right) \left(-\frac{1}{2}\alpha'\right) K^{\mu} \\ & \quad + \partial_a x^{\mu} + : \partial_a x^{\mu} \sum_{n=2,1}^{\infty} \end{aligned}$$

So, since we are only counting terms with a single contraction we find, since $j^{\mu} = \frac{i}{\alpha'} \partial_a x^{\mu}$ $\frac{i}{\alpha'} \alpha' (-1/2) = \frac{1}{2}$

$$j^{\mu}(z) : e^{ikX(0,0)} : \sim \frac{K^{\mu}}{2z} : e^{ikX(0,0)} :$$

$$\text{Similarly: } \bar{j}^{\mu}(\bar{z}) : e^{ikX(0,0)} : \sim \frac{K^{\mu}}{2\bar{z}} : e^{ikX(0,0)} :$$

* For the world-sheet translations we have $\partial\sigma^a \in V^a$

$\therefore \delta X^m = \frac{\delta \sigma^a}{\delta \sigma^a} \delta \sigma^a = \partial_a X^m \epsilon V^a$
 (Book has $\delta_{WS} X^m = -\epsilon V^a \partial_a X^m$, not sure why theres a minus sign).

$$S = \frac{1}{2\pi\alpha'} \int d^2\sigma \partial^a X_m \partial_a X^m$$

$$\delta_{WS} S = \frac{1}{2\pi\alpha'} \int d^2\sigma \partial^a X_m \partial_a (-\epsilon V^b \partial_b X^m)$$

$$= \frac{1}{2\pi\alpha'} \int d^2\sigma \partial^a X_m [\partial_a (\epsilon V^b) \partial_b X^m + \epsilon V^b \partial_a \partial_b X^m]$$

$$\delta_{WS} S = \frac{1}{2\pi\alpha'} \int d^2\sigma \partial^a X_m \partial_a (\epsilon V^b) \partial_b X^m - \frac{1}{2\pi\alpha'} \int d^2\sigma \epsilon V^b \partial_a \partial_b X_m \partial^a X^m$$

Switch $a \leftrightarrow b$

$$+ \frac{1}{2\pi\alpha'} \int d^2\sigma (\partial_b \epsilon) V^b \partial_a X^m \partial_a X_m + \dots$$

$$\delta_{WS} S = \frac{1}{2\pi\alpha'} \int d^2\sigma (\partial_a \epsilon) \left[\partial^a X_m \partial_b X^m + V^a \partial_b X^m \partial^b X_m \right]$$

...

$$J_a = i V^b T_{ab}$$

$$T_{ab} = -\frac{1}{\alpha'} : (\partial_a X^m \partial_b X_m - \frac{1}{2} \delta_{ab} \partial_c X^m \partial^c X_m) :$$

Conformal Invariance: Note

$$T_{ab} \eta^{ab} = -\frac{1}{\alpha'} : \partial_a X^m \partial^a X_m - \partial_c X^m \partial^c X_m : = 0$$

i.e. T_{ab} is traceless. $T^a_a = 0$

In $\mathbb{D}$ coords:

$$T_{\bar{z}\bar{z}} = 0$$

$$\partial^a T_{ab} = 0 \Leftrightarrow \bar{\partial} T_{z\bar{z}} = \partial T_{\bar{z}\bar{z}} = 0$$

$$\Rightarrow T(z) \equiv T_{zz}(z) \quad \bar{T}(\bar{z}) \equiv T_{\bar{z}\bar{z}}(\bar{z})$$

$$T(z) = -\frac{1}{\alpha'} : \partial X^\mu \partial X_\mu :$$

$$\bar{T}(\bar{z}) = -\frac{1}{\alpha'} : \bar{\partial} X^\mu \bar{\partial} X_\mu :$$

$$j(z) = i v(z) T(z), \quad \bar{j}(\bar{z}) = i v(z)^* \bar{T}(\bar{z})$$

are conserved for $h_{01} \neq 0 \Rightarrow v(z)$.

$$T_{zz} \equiv T(z) = -\frac{1}{\alpha'} : \partial X^\mu \partial X_\mu - \frac{1}{2} \partial^a X_\mu \partial_a X^\mu :$$

where

$$\delta_{zz} = 0$$

$$:T(z) X^\mu(0): = -\frac{1}{\alpha'} : \partial X^\mu \partial X_\mu : : X^\mu(0) :$$

$$= : \partial X^\mu \partial X_\mu : X^\mu(0)$$

